
Compositional Generation in Conditional Diffusion Models: The Effect of Factor Correlation on Compositional CFG

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Marthe Elgawly - 2170201

Abstract

Compositional generation combines single factor scores assuming independent factors. Using a class-conditional DDPM on dSprites, this work tests training correlation ($\rho \in \{0.0, 0.25, 0.50, 0.75, 0.95\}$). Minority accuracy drops from 0.63 to 0.32 for joint conditioning and from 0.30 to 0.17 for naive composition. Weight calibration gives no consistent improvement, while entanglement increases with ρ (0.82 vs. 0.33 baseline). Under true zero-shot exclusion, composition instead outperforms joint conditioning (0.125 vs. 0.059), showing its value when joint training data is unavailable.

1. Introduction

Diffusion models generate images through iterative denoising (Ho et al., 2020; Croitoru et al., 2023). Classifier-Free Guidance enables conditional generation, while compositional generation combines single-factor scores at inference without joint training (Ho & Salimans, 2022; Liu et al., 2023). Both assume conditional independence, which may fail under correlated factors. This work studies this effect using controlled correlations and a zero-shot split that removes minority combinations. Joint and compositional generation are evaluated with a factor-recovery classifier, while entanglement and weight-calibration analyses examine attribute leakage and guidance miscalibration.

2. Related Work

DDPM (Ho et al., 2020) introduced noise-prediction training with U-Nets (Ronneberger et al., 2015). Classifier-Free Guidance enables conditional control via label dropout, while compositional methods combine separately conditioned scores at inference (Ho & Salimans, 2022; Liu et al.,

2023; Du et al., 2023). DDIM (Song et al., 2020) reduces sampling cost through deterministic sampling. Prior work studies compositional emergence (Okawa et al., 2023) and data withholding (Jeong et al., 2026), but does not directly control factor correlation. Correlated training can distort learned marginals (Gaudi et al., 2025). This work instead varies correlation strength directly to isolate its effect on compositional score estimates.

3. Method

3.1. Dataset and preprocessing

The dSprites dataset (Matthey et al., 2017) contains 737,280 procedural 64×64 binary images spanning the Cartesian product of six independent ground-truth latent factors, serving as a controlled testbed for evaluating latent disentanglement and compositional generalization. For all correlated and zero-shot subsets, only the fixed white-color subset was used, with reduced factor ranges, cropped position and orientation, and scale coarsened into $n_{\text{scale}} = 3$ bins (Appendix: Figure 1).

3.2. Model Architecture

The conditional U-Net ($\sim 22.8\text{M}$ parameters, base width 96) spans resolutions 64, 32, 16, 8 with skip connections. It uses distinct shape and scale embedding branches—including null tokens—injecting summed sinusoidal timestep and factor embeddings into each residual block.

3.3. Training Setup

For each ρ , models are trained for 200 epochs via AdamW ($\text{lr} = 2 \times 10^{-4}$, cosine annealing, batch size 256, FP16) on a Tesla T4. The DDPM uses $T = 300$ steps with MSE noise prediction (Eq. 1).

$$\mathcal{L} = \mathbb{E}_{t, \mathbf{x}_0, \epsilon} [\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t, c_{\text{shape}}, c_{\text{scale}})\|^2]. \quad (1)$$

Shape and scale labels are dropped independently ($p_{\text{drop}} = 0.2$) for joint, single-factor, and unconditional sampling.

Email: Marthe Elgawly <elgawly.2170201@studenti.uniroma1.it>.

3.4. Correlated Training Subsets

For each ρ , a balanced 6,000- subset sets $P(\text{scale} \mid \text{shape})$ from uniform ($\rho = 0$) to near-deterministic ($\rho \rightarrow 1$) per shape. The six minority combinations remain present, not fully excluded, at all swept ρ . Full exclusion is evaluated separately (Section 4.3).

3.5. Sampling: Joint vs. Composed

Joint sampling queries both factors simultaneously.

Composed sampling (Liu et al., 2023) combines null, shape-only, and scale-only predictions as shown in Equation 2.

$$\epsilon_{\text{comp}} = \epsilon_{\text{null}} + w_{\text{shape}}(\epsilon_{\text{shape}} - \epsilon_{\text{null}}) + w_{\text{scale}}(\epsilon_{\text{scale}} - \epsilon_{\text{null}}) \quad (2)$$

with $w_{\text{shape}} = w_{\text{scale}} = 1$, using DDIM at 50 steps – refer to Section 4.2, which checks that this does not under-sample relative to more expensive alternatives.

3.6. Evaluation

An evaluation 3-layer CNN (32/64/128 channels; Adam, $\text{lr} = 10^{-3}$, 10 epochs, $n = 3121$) shows uniform training convergence across all ρ (Appendix Figure 7), isolating model behavior from optimization artifacts. **Factor-recovery accuracy** measures shape and scale recovery on target combinations (real-image baseline: shape acc. 0.949, scale acc. 0.990, $n = 3121$). **FFD** evaluates feature similarity in the classifier’s 128-D embedding space, replacing FID for binary silhouettes and small sample sizes (lower is better). **Nearest-neighbor distance** measures pixel-space L_2 distance to the closest training image.

4. Results

4.1. Compositional Degradation Across Correlation

Composed accuracy lags behind joint accuracy across all ρ , falling from 0.297 ± 0.016 to 0.166 ± 0.013 versus 0.633 ± 0.017 to 0.322 ± 0.017 (Appendix Table 2, Figure 4). Because in-distribution accuracy tops out below 0.65, this cannot stem only from correlation-induced compositional failure.

4.2. Ablation, Weight Calibration, and Entanglement

Factor-recovery accuracy is robust to DDIM step counts (0.688–0.750 across fixed-noise reruns; Appendix Table 1), validating the 50-step default, while weight sweeps identify $w = 1.0$ as near-optimal at 4 of 5 ρ (Appendix Table 3). Factor leakage is bidirectional yet asymmetric: conditioning on shape 0 monotonically leaks scale ($0.35 \rightarrow 0.82$), and scale 2 strongly leaks shape 2 ($0.41 \rightarrow 0.79$), whereas other pairs show weak, non-monotonic trends (Appendix

Table 4).

4.3. True Zero-Shot Generalization

Under this true zero-shot benchmark—which excludes rather than correlates minority combinations—composed sampling more than doubles joint accuracy (0.125 vs. 0.059; Figures 2–3) by generating recognizable, scaled shapes instead of blank or noisy failures, proving composition succeeds where joint conditioning fails to generalize.

4.4. Second-Seed Robustness

A second seed (Appendix Table 7) confirms composed accuracy consistently lags behind joint accuracy, showing qualitative rather than numerical robustness despite large value shifts (e.g., $\rho = 0$ joint accuracy falls from 0.633 to 0.375). FFD and nearest-neighbor metrics (Appendix Table 6) rule out memorization across all ρ . Finally, classifier guidance sweeps failed to outperform naive composition (best zero-shot: 0.000; best $\rho = 0.95$: 0.016), confirming the compositional gap is representational rather than hyperparameter-driven.

5. Discussion and Conclusions

Across seeds and ρ , composed accuracy consistently trails joint accuracy despite seed-level variation (Appendix Table 7). With recalibration and DDIM step artifacts ruled out, representation entanglement remains the primary culprit, while with an unexplained asymmetry: shape 0 exhibits clean monotonic leakage, whereas shapes 1–2 show weaker, non-monotonic trends. Under zero-shot exclusion, this dynamic inverts: composition outperforms joint sampling. While correlated rarity induces entanglement, total exclusion forces blind joint interpolation, allowing composition to exploit isolated single-factor cues. Compositional CFG depends on the data-omission regime rather than mere rarity. Finally, classifier guidance sweeps failed to close this gap, indicating an inherent representational bottleneck.

5.1. Limitations and Future Work

Findings are limited to 3 shapes, 3 scale bins, 64×64 resolution, one architecture, and a single noise schedule, leaving scaling to richer factor spaces untested. Aggregate training loss logging prevents per-factor convergence analysis, while small samples ($n = 256$ –3,000) warrant a larger-scale check of the shape-0 entanglement asymmetry. Running only two seeds leaves endpoint variance ambiguity unresolved. Additionally, the guidance classifier was trained solely on clean images rather than noisy latents x_t . Cosine schedules, larger architectures, and real-world correlated attributes remain possible extensions.

6. Statement on the Use of AI

AI tools (Gemini, Claude and DeepSeek) were used as a consulting tool in order to boost the idea into a code that is heavy and deeply targets a gap in the studied paper. AI was used to check whether the chosen topic matches the required guidelines or not - given the accepted topics. In terms of coding, AI was used for debugging, fixing codes that encountered errors along the way, plotting codes (not plotting the actual images but providing me with the code to plot the graphs needed). AI was used to add codes to reload saved runs so that i can add some extra codes onto it. AI was used to explain theoretical concepts from the reference papers followed. It also was used to understand some outputs throughout the coding process so that i can know if the output is as expected or not. AI was also used as guidance when writing the report to cover in each section with what level of detail to keep text concise and within the page limit. It was used at the end as a grammar check, text flow check, and to check whether we covered everything from our code or missed some details as well as to summarize the text to fit the page limit. AI was used to generate the equations in latex format for the report and code (markdowns added in the code) All submitted code has been fully run, verified, understood by me. All results, figures, conclusions, discussions mine. AI was not used to generate any of the figures or the report text but rather fix the english used.

Note: All referenced images and tables are included in the appendix.

References

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Appendix

One real dSprites example per (shape, scale) combination

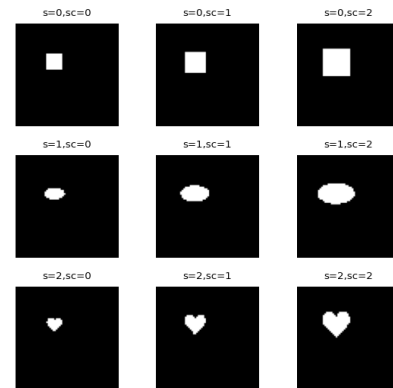


Figure 1. One real dSprites example per (shape, scale) combination after preprocessing.

¹ $w=1.5$ is within noise at 0.145.

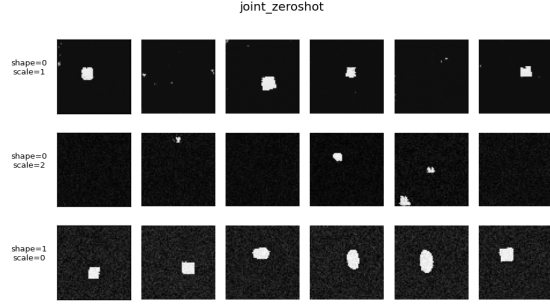


Figure 2. Joint sampling under true zero-shot exclusion: shape-0 and shape-1 minority combinations. Outputs seem blank or noise-like.

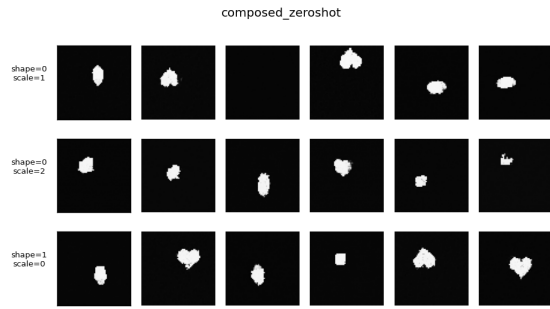


Figure 3. Composed sampling under true zero-shot exclusion, using the same combinations as Figure 2. Outputs show a scaled, located object, but not always the requested shape identity.

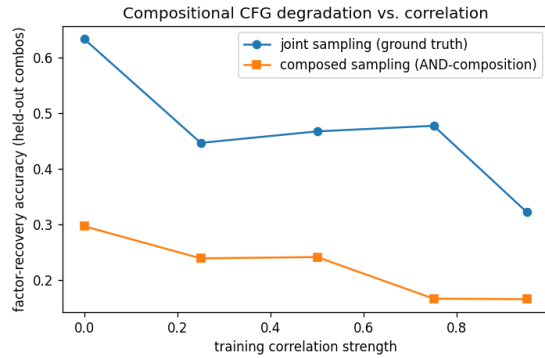


Figure 4. Joint vs. naive composed accuracy on minority combinations across correlation levels.

Steps	Shape acc.	Scale acc.	Joint acc.
50	0.781	0.938	0.734
100	0.875	0.969	0.859
200	0.641	1.000	0.641
300	0.703	0.969	0.688

Table 1. Factor-recovery accuracy vs. DDIM step count, $\rho = 0$, one in-distribution combination, $n = 64$.

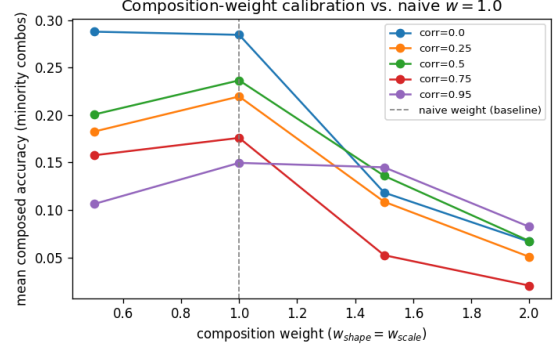


Figure 5. Composed accuracy vs. composition weight, per correlation level.

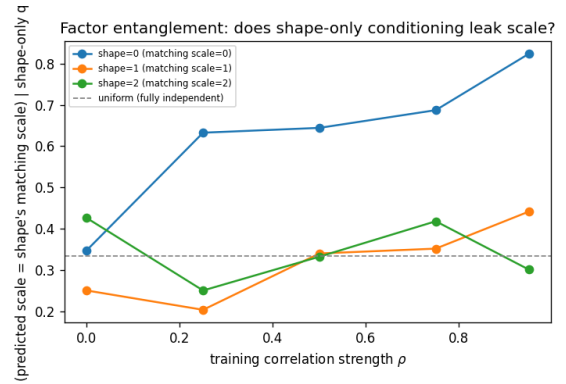


Figure 6. Predicted probability of the shape's matching scale under shape-only conditioning, across correlation levels.

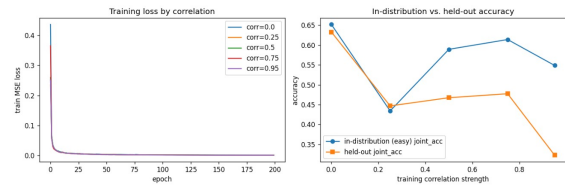


Figure 7. Left: training MSE loss by correlation level, all converging to comparable final loss (~ 0.001 – 0.002). Right: in-distribution vs. held-out (minority) joint accuracy across correlation levels.

ρ	In-dist. joint acc.	Minority joint acc.	Minority composed acc.
0.00	0.652	0.633	0.297
0.25	0.434	0.447	0.239
0.50	0.589	0.467	0.242
0.75	0.614	0.477	0.167
0.95	0.549	0.322	0.166

Table 2. Mean accuracy across the six minority combinations, and the in-distribution sanity-check accuracy, at each correlation level.

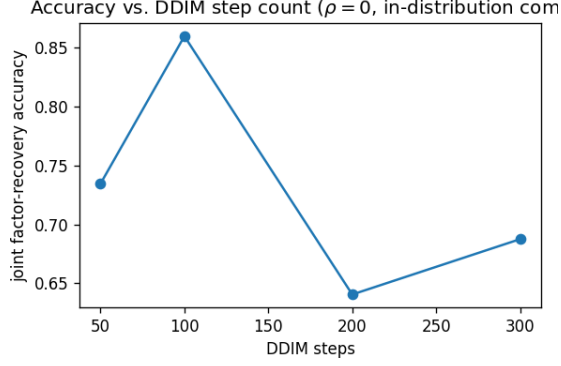


Figure 8. Joint factor-recovery accuracy vs. DDIM step count at $\rho = 0$.

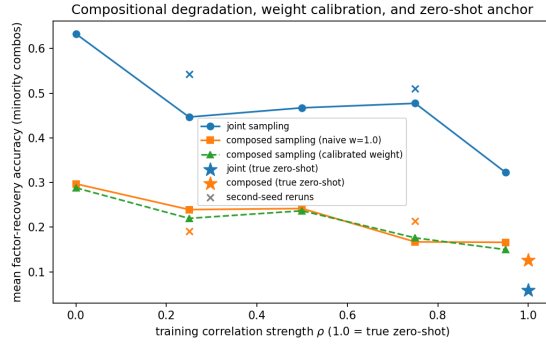


Figure 9. Combined view: degradation curve, weight-calibrated composition, true zero-shot anchor, and second-seed reruns across all correlation levels.

ρ	Best w (Acc.)	Naive $w=1.0$ Acc.
0.00	0.5 (0.288)	0.285
0.25	1.0 (0.220)	0.220
0.50	1.0 (0.237)	0.237
0.75	1.0 (0.176)	0.176
0.95	1.0 (0.150) ¹	0.150

Table 3. Best composed accuracy by weight sweep vs. naive baseline.

ρ	Shape 0 ($s=0$)	Shape 1 ($s=1$)	Shape 2 ($s=2$)
0.00	0.348	0.250	0.426
0.25	0.633	0.203	0.250
0.50	0.645	0.340	0.332
0.75	0.688	0.352	0.418
0.95	0.824	0.441	0.301

Table 4. $P(\text{predicted scale} = \text{matching scale})$ for shape-only queries across ρ .

Sampling Method	Mean Minority Acc.
Joint (zero-shot)	0.059
Composed (zero-shot)	0.125

Table 5. True zero-shot accuracy: joint vs. composed sampling.

ρ	Joint FFD	Comp. FFD	NN Mean	NN Min
0.00	137.91	67.01	16.13	4.86
0.25	279.79	570.52	12.60	3.75
0.50	246.22	317.05	14.38	6.10
0.75	246.90	240.40	16.27	4.35
0.95	413.18	866.56	12.72	4.32

Table 6. Fréchet Factor Distance and nearest-neighbor distance across correlation levels.

ρ	Joint Acc. (Seed 2)	Comp. Acc. (Seed 2)
0.00	0.375	0.217
0.25	0.543	0.191
0.50	0.347	0.195
0.75	0.510	0.214
0.95	0.443	0.147

Table 7. Second-seed verification at all five correlation levels.